Exploring Daily Weather Patterns in State College, Pennsylvania

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1 Introduction

This report explores daily weather patterns in State College, Pennsylvania using daily station-level weather records from NOAA Climate Data Online. The purpose of the analysis is to understand how local weather varies across years, months, and seasons. The project focuses on maximum temperature, minimum temperature, precipitation, snowfall, and snow depth.

The main research question is: **What patterns can we observe in daily weather in State College, Pennsylvania over the last ten years?** More specifically, the analysis examines whether State College shows clear seasonal temperature patterns, which months tend to receive more precipitation, and how snowfall varies across years.

2 Data Provenance

The data used in this report come from NOAA Climate Data Online, specifically the Global Historical Climatology Network Daily data for the State College, Pennsylvania weather station. The station used for this project is **STATE COLLEGE, PA US**, station ID **GHCND:USC00368449**. The working file for this analysis is `SC_weather_10.csv`.

The dataset contains daily observations. Important original variables include daily maximum temperature (`TMAX`), daily minimum temperature (`TMIN`), precipitation (`PRCP`), snowfall (`SNOW`), and snow depth (`SNWD`). Additional variables were created during data cleaning, including `year`, `month`, `season`, `temp_range`, `rain_day`, `snow_day`, `heavy_prcp_day`, and `freezing_day`.

Because the data come from NOAA, a public scientific data provider, the dataset is appropriate for reproducible exploratory data analysis. However, the analysis should still be interpreted carefully because 2016 and 2026 are partial years in the current file, rather than complete calendar years.

3 FAIR and CARE Principles

The dataset generally satisfies the **FAIR principles**. It is findable because it can be located through NOAA Climate Data Online using the station ID. It is accessible because NOAA provides public access to daily weather station data. It is interoperable because the data are stored in a standard CSV format and use common weather variables. It is reusable because the data include clear variable names, station information, and measurement records that can be analyzed in R.

The **CARE principles** are less directly applicable because this project does not use Indigenous data or community-owned cultural data. However, the project still follows the spirit of responsible data use by clearly describing the source, avoiding unsupported causal claims, and using the data only for exploratory weather analysis. The report also acknowledges NOAA as

the original data provider and maintains transparency about created variables and partial-year observations.

4 Exploratory Data Analysis

4.1 Annual Weather Summary

Table 1 summarizes the annual weather records. Complete years generally contain 365 or 366 days, while 2016 and 2026 are partial years in this file. For that reason, comparisons involving total precipitation or total snowfall should focus mainly on complete years.

Table 1: Annual weather summary for State College, PA.

Year	Days observed	Avg. Maximum Temp	Avg. Minimum Temp	Avg. Daily Temp Range	Total Precipitation	Total Snowfall	Max Snow Depth	Rain Days	Snow Days	Freezing Days
2016	256	67.6	50.5	17.1	24.89	7.0	4	102	7	36
2017	365	59.4	43.3	16.2	42.19	37.7	10	161	23	114
2018	365	57.9	42.8	15.1	63.75	39.1	11	184	24	131
2019	365	59.0	42.4	16.6	41.50	32.5	7	155	23	128
2020	366	60.9	43.9	17.0	39.10	25.2	13	151	19	110
2021	365	61.7	44.0	17.7	45.94	32.3	12	157	31	121
2022	365	60.8	41.7	19.1	39.80	43.0	5	147	26	130
2023	365	61.9	44.0	18.0	39.97	11.1	2	154	16	102
2024	366	61.6	44.4	17.2	43.18	28.0	5	144	19	100
2025	365	59.6	42.3	17.3	38.36	21.8	4	142	28	122
2026	110	45.0	27.2	17.9	9.17	20.5	10	40	11	67

The table shows that annual average maximum temperatures for complete years are mostly near the upper 50s to low 60s in degrees Fahrenheit. The largest annual precipitation total appears in 2018, while 2022 has the largest total snowfall among the complete years shown.

4.2 Annual Temperature Pattern

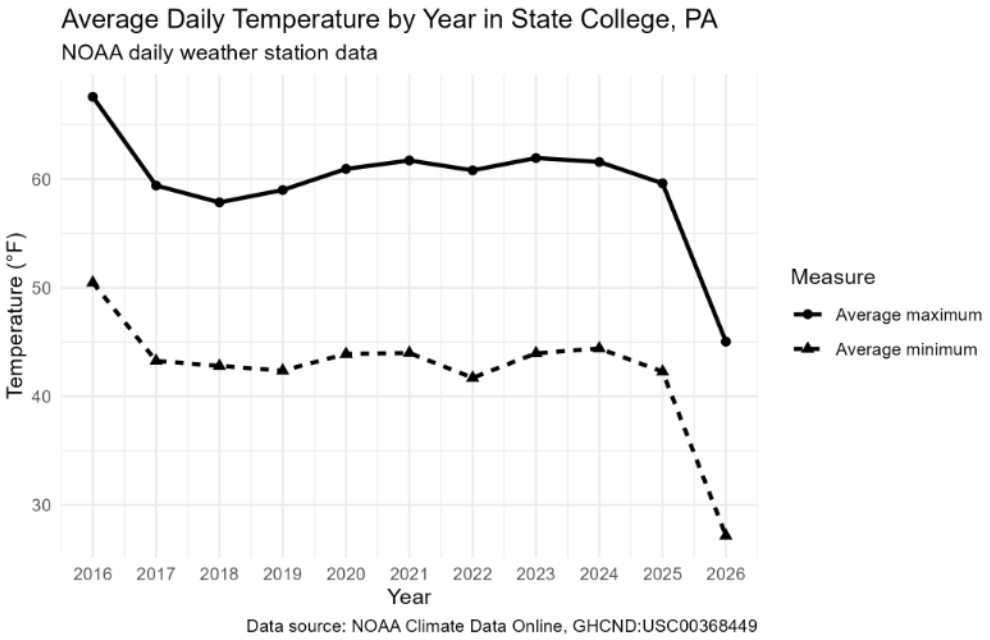

Figure 1: Line graph showing average annual maximum and minimum temperatures in State College, Pennsylvania. The two lines show that complete-year average maximum temperatures usually stay near the upper 50s to low 60s, while average minimum temperatures usually stay in the low to mid 40s. The partial year 2026 appears much colder because it does not include a full calendar year.

Figure 1 shows that the annual temperature pattern is relatively stable for complete years. Average maximum temperature usually remains close to 60°F, while average minimum temperature is generally in the low to mid 40s. The sharp drop in 2026 should not be interpreted as a long-term cooling pattern because 2026 is only a partial year in the dataset.

4.3 Monthly Weather Summary

Table 2 compares monthly averages and totals across the full dataset. This table helps show the seasonal structure of weather in State College.

Table 2: Monthly weather summary for State College, PA.

Month	Avg. Maximum Temp	Avg. Minimum Temp	Median Daily Precipitation	Total Precipitation	Total Snowfall	Rain Days	Snow Days
January	35.4	22.1	0.00	28.07	88.9	137	65
February	40.5	24.0	0.00	29.08	76.4	113	58
March	49.0	30.3	0.00	30.71	44.3	134	33
April	60.5	41.1	0.01	36.72	9.6	159	6
May	69.1	51.2	0.01	48.86	0.0	160	0
June	77.7	59.7	0.00	38.98	0.0	125	0
July	83.3	65.3	0.00	42.65	0.0	127	0
August	80.6	63.0	0.00	43.21	0.0	124	0
September	74.3	55.9	0.00	37.83	0.0	108	0
October	63.6	45.1	0.00	34.70	0.0	119	0
November	49.8	33.2	0.00	26.01	21.9	97	18
December	39.5	26.5	0.00	31.03	57.1	134	47

The monthly table shows a clear seasonal temperature cycle. July has the highest average maximum temperature, while January has the lowest average maximum and minimum temperatures. Snowfall is concentrated in winter and early spring months, especially January, February, March, and December.

4.4 Monthly Precipitation

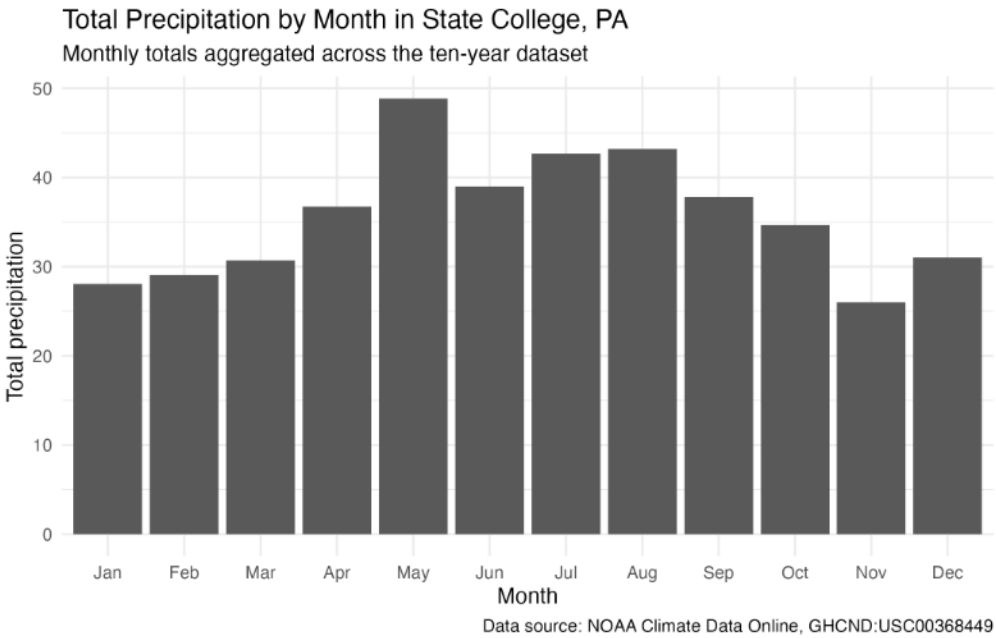

Figure 2: Bar chart showing total precipitation by month in State College, Pennsylvania. May has the highest total precipitation, while November has one of the lowest totals. Precipitation occurs in all months, but the total amount varies by season.

Figure 2 shows that precipitation is not limited to one season. May has the highest total precipitation in the summarized dataset, while late fall and winter months tend to be lower than spring and summer months. This supports the idea that rainfall and precipitation in State College occur throughout the year but vary by month.

4.5 Distribution of Daily Precipitation

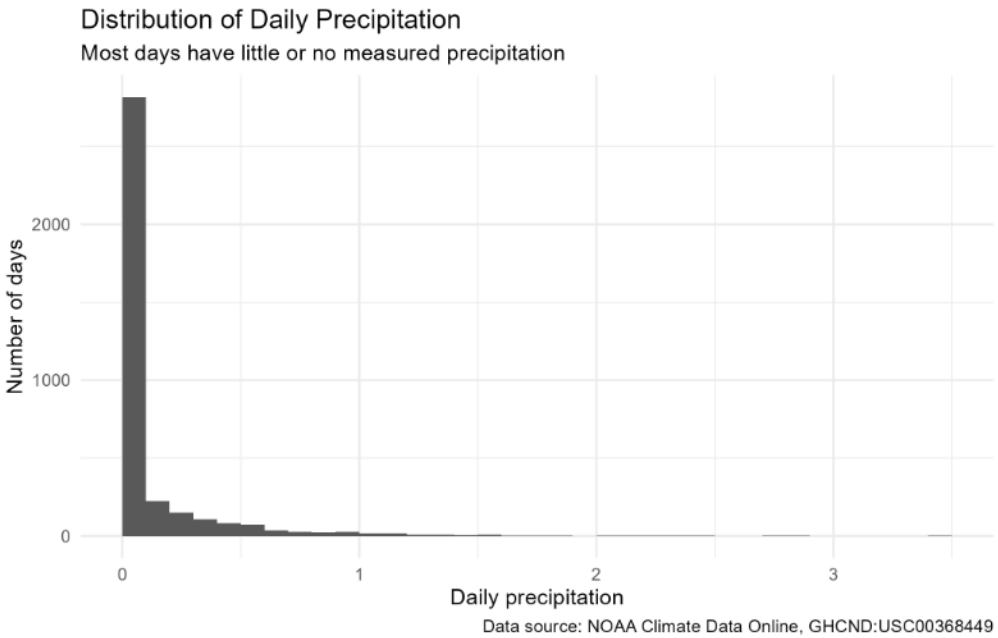

Figure 3: Histogram showing the distribution of daily precipitation. Most observations are near zero, with a long right tail for days with heavier precipitation.

Figure 3 shows that daily precipitation is strongly right-skewed. Most days have little or no precipitation, while a small number of days have much larger precipitation values. This explains why the median daily precipitation in many months is 0.00 even though the monthly total precipitation is not zero.

4.6 Seasonal Temperature Distribution

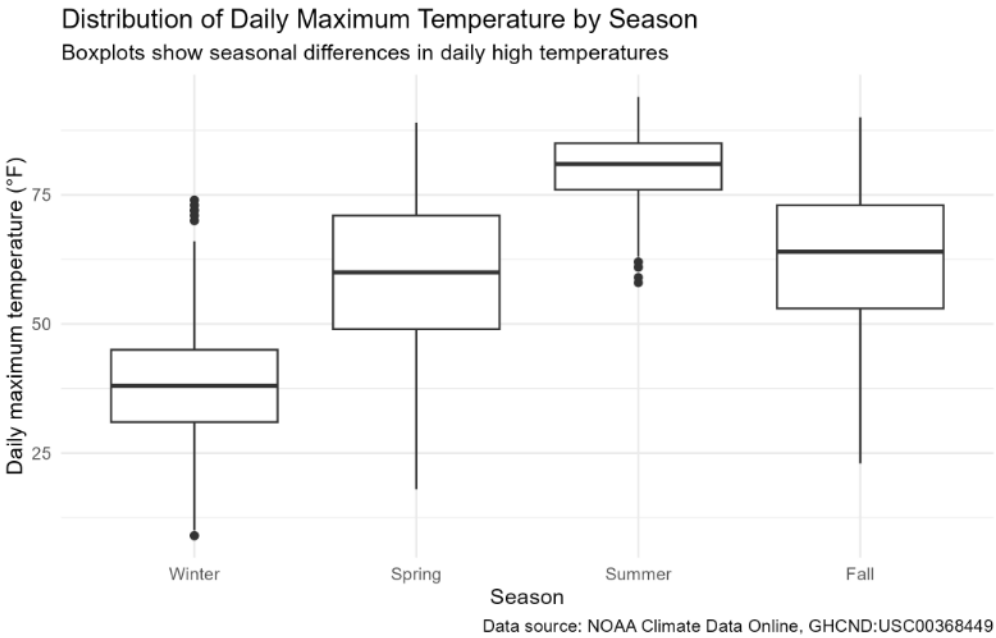

Figure 4: Boxplots showing the distribution of daily maximum temperature by season. Summer has the highest median and the narrowest warm-season range, while winter has the lowest median and several warmer outliers. Spring and fall show wider transition-season variation.

Figure 4 shows that daily maximum temperatures differ strongly by season. Summer has the highest median daily maximum temperature, while winter has the lowest. Spring and fall show broader variation because they are transition seasons.

4.7 Annual Snowfall

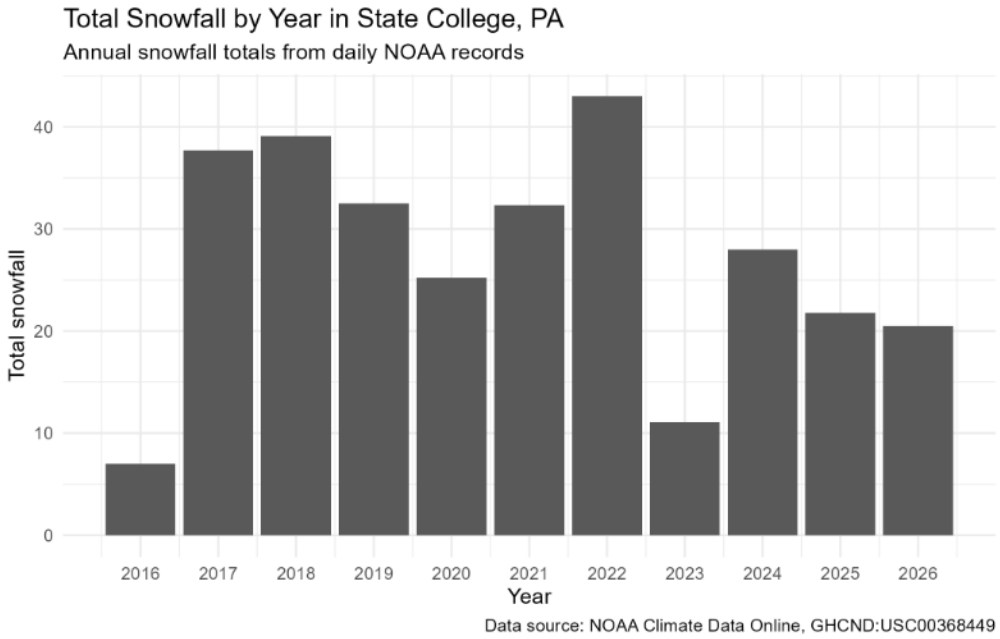

Figure 5: Bar chart showing total snowfall by year in State College, Pennsylvania. Snowfall varies substantially by year, with 2022 among the highest complete years and 2023 among the lowest complete years.

Figure 5 shows that snowfall varies substantially from year to year. Among the complete years, 2022 has the highest total snowfall, while 2023 has a much lower snowfall total. This suggests that winter weather in State College can vary considerably even when annual average temperatures are relatively stable.

5 Overall Data Story

The exploratory analysis shows that State College has a strong seasonal weather pattern. Temperatures are lowest in winter, rise through spring, peak in summer, and decline through fall. Precipitation occurs throughout the year, but the monthly totals suggest that spring and summer months can contribute heavily to annual precipitation. Snowfall is concentrated in winter and early spring, but the total amount of snowfall varies considerably across years.

The analysis also highlights the importance of distinguishing daily distributions from monthly or annual totals. For example, many days have no precipitation, but enough rainy days and

heavier precipitation events accumulate to create meaningful monthly totals. Similarly, annual averages can look stable while snowfall varies strongly across years.

The partial-year observations for 2016 and 2026 are important context. These years should not be compared directly with full years when interpreting total precipitation, total snowfall, or annual average temperature.

6 Open Science and Reproducibility

This project follows open science principles by using a public data source, documenting the data provenance, keeping analysis code in a reproducible R workflow, and reporting created variables clearly. The final project repository should include the raw CSV file, the QMD report, generated figures, citation files, and any scripts used to clean and analyze the data.

The project also applies PCIP by making the report readable for a broad audience, using clear captions and alt text, and avoiding unsupported claims. The analysis is exploratory rather than causal, so the report focuses on describing patterns rather than claiming that one weather pattern causes another.

7 Author Contributions

- **Ata:** Contributed to project planning, data interpretation, and review of weather summaries.
- **Jincheng:** Contributed to table review, figure interpretation, and report organization.
- **Qihaohan:** Primary author for the R data cleaning and analysis workflow; contributed to figure generation and report writing.

8 Code Appendix

The following code was used to read, clean, summarize, and visualize the data. Code is placed in the appendix so that the main body of the report focuses on interpretation.